

ST. XAVIER'S INTER COLLEGE. MARIAHU, JAUNPUR
Pre – Boards (2025) Affiliated to C.B.S.E. (New Delhi)
CLASS- 10TH SUBJECT – MATHS MM.80

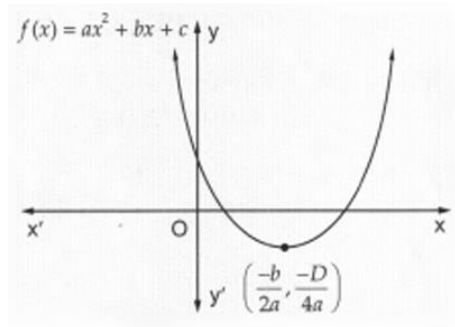
NOTE – Attempt all the question

Section A

1. (HCF $\times$ LCM) for the numbers 30 and 70 is:

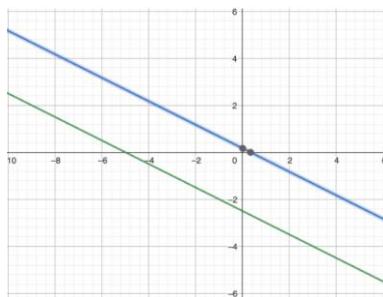
- a) 21 b) 70
 c) 2100 d) 210

2. Figure show the graph of the polynomial $f(x) = ax^2 + bx + c$ for which



- a) $a > 0$, $b < 0$ and $c > 0$ b) $a < 0$, $b < 0$ and $c < 0$
 c) $a < 0$, $b > 0$ and $c > 0$ d) $a > 0$, $b > 0$ and $c < 0$

3. The pair of equations $x + 2y + 5 = 0$ and $-3x - 6y + 1 = 0$ have



- a) a unique solution b) infinitely many solutions
 c) no solution d) exactly two solutions

4. In the Maths Olympiad of 2020 at Animal Planet, two representatives from the donkey's side, while solving a quadratic equation, committed the following mistakes.

- i. One of them made a mistake in the constant term and got the roots as 5 and 9.
 ii. Another one committed an error in the coefficient of x and he got the roots as 12 and

But in the meantime, they realised that they are wrong and they managed to get it right jointly. Find the quadratic equation.

a) $2x^2 + 7x - 24 = 0$

b) $x^2 + 4x + 14 = 0$

c) $3x^2 - 17x + 52 = 0$

d) $x^2 - 14x + 48 = 0$

5. What is the common difference of an AP in which $818 - 814 = 32$?

a) -8

b) 4

c) -4

d) 8

6. ABCD is a rectangle whose three vertices are B (4,0), C (4,3) and D (0,3). The length of one of its diagonals is

a) 5

b) 3

c) 4

d) 25

7. In what ratio, does x-axis divide the line segment joining the points A(3, 6) and B(-12, -3)?

a) 2:1

b) 1:2

c) 4:1

d) 1:4

8) Given below is the picture of the Olympic rings made by taking five congruent circles of radius 1cm each, intersecting in such a way that the chord formed by joining the point of intersection of two circles is also of length 1cm. Total area of all the dotted regions (assuming the thickness of the rings to be negligible) is



(A) $4[\pi - \sqrt{3}] \text{ cm}^2$

(B) $[\pi - \sqrt{3}] \text{ cm}^2$

(C) $4[\pi - \sqrt{3}] \text{ cm}^2$

(D) $8[\pi - \sqrt{3}] \text{ cm}^2$

9) A pair of dice is tossed. The probability of not getting the sum eight is

(A) $5/36$

(B) $31/36$

(C) $5/18$

(D) $5/9$

10) The sum of two numbers is 1215 and their HCF is 81, then the possible pairs of such numbers are

(A) 2

(B) 3

(C) 4

(D) 5

- 11) If zeroes of the quadratic polynomial $ax^2 + bx + c$ ($a, c \neq 0$) are equal, then
 (A) c and b must have opposite signs (B) c and a must have opposite signs
 (C) c and b must have same signs (D) c and a must have same signs
- 12) The area (in cm^2) of a sector of a circle of radius 21cm cut off by an arc of length 22cm is
 (A) 441 (B) 321
 (C) 231 (D) 221
- 13) The points A(9,0), B(9, -6), C(-9,0) and D(-9,6) are the vertices of a
 (A) Square (B) Rectangle
 (C) Parallelogram (D) Trapezium
- 14) If $\triangle ABC \sim \triangle DEF$, $AB=6\text{cm}$, $DE=9\text{cm}$, $EF=6\text{cm}$ and $FD=12\text{cm}$, then the perimeter of $\triangle ABC$ is
 (A) 28cm (B) 28.5cm
 (C) 18cm (D) 23cm
- 15) If $a = 2 \times 2 \times 3 \times x$, $b = 2 \times 2 \times 3 \times 5$, $c = 2 \times 2 \times 3 \times 7$ and $\text{LCM}(a, b, c) = 3780$, then x is equal to
 (A) 1 (B) 2
 (C) 3 (D) 0
- 16) The shortest distance (in units) of the point (2,3) from y-axis is
 (A) 2 (B) 3
 (C) 5 (D) 1
- 17) The points A(9,0), B(9, -6), C(-9,0) and D(-9,6) are the vertices of a
 (A) Square (B) Rectangle
 (C) Parallelogram (D) Trapezium
- 18) $1 + \frac{\cot^2 \alpha}{1 + \sec \alpha} =$
 a) $\sin \alpha$ b) $\sec \alpha$
 c) $\cos \alpha$ d) $\tan \alpha$

DIRECTIONS: In the question number 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option:

- (A) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A)
 (B) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A)

(C) Assertion (A) is true but reason (R) is false.

(D) Assertion (A) is false but reason (R) is true.

19) **Assertion (A):** The number $5n$ cannot end with the digit 0, where n is a natural number

Reason (R): A number ends with 0, if its prime factorization contains both 2 and 5

20) **Assertion (A):** If $\cos A + \cos 2A = 1$, then $\sin 2A + \sin 4A = 1$

Reason (R): $\sin 2A + \cos 2A = 1$

(Section – B)

Section B consists of 5 questions of 2 marks each.

21) (A) The A.P 8, 10, 12, has 60 terms. Find the sum of last 10 terms.

OR

(B) Find the middle term of A.P 6, 13, 20,, 230

22) If $\sin(A + B) = 1$ and $\cos(A - B) = \frac{\sqrt{3}}{2}$, $0^\circ < A, B < 90^\circ$, find the measure of angles A and B .

23) Show that: $\tan^2 \theta + \tan^2 \theta = \sec^2 \theta - \sec^2 \theta$ for $0^\circ \leq \theta \leq 90^\circ$

OR

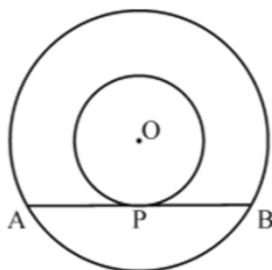
$\tan \theta$ Prove that: $1 - \cot \theta + \cot \theta - \tan \theta = 1 + \tan \theta + \cot \theta$

24) Find the length of the arc of a circle of diameter 42 cm which subtends an angle of 60° at the centre.

OR

A horse is tethered to one corner of a rectangular field of dimensions $70 \text{ m} \times 52 \text{ m}$, by a rope of length 21 m. How much area of the field can it graze?

25) Two concentric circles with centre O are of radii 3 cm and 5 cm. Find the length of chord AB of the larger circle which touches the smaller circle at P



(Section – C)

26) Prove that $3 + \sqrt{5}$ is an irrational number

27) If α and β are the zeros of the polynomial $f(x) = 6x^2 + x - 2$, find the value of $\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)$

28) A plane left 30 minutes late than its scheduled time and in order to reach the destination 1500 km away in time, it had to increase its speed by 100 km/h from the usual speed. Find its usual speed.

29) prove that

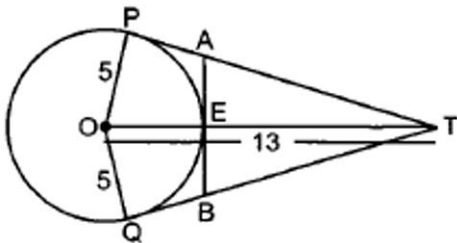
$$\frac{\tan \theta + \sec \theta - 1}{\tan \theta - \sec \theta + 1} = \frac{1 + \sin \theta}{\cos \theta}$$

30) The percentage of marks obtained by 100 students in an examination are given below:

Marks	30-35	35-40	40-45	45-50	50-55	55-60	60-65
Frequency	14	16	18	23	18	8	3

Determine the median percentage of marks

31) In figure, O is the centre of a circle of radius 5 cm. T is a point such that OT = 13 cm and OT intersect circle at E. If AB is a tangent to the circle at E, find the length of AB. where TP and TQ are two tangents to the circle

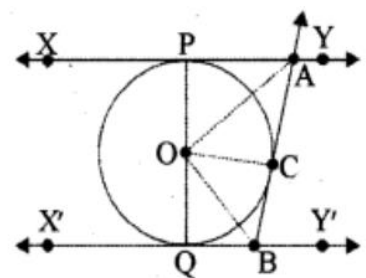


(Section-D)

32) solve $\frac{x-1}{2x+1} + \frac{2x+1}{x-1} = 2, x \neq -\frac{1}{2}, 1$

33) ₹ 9000 were divided equally among a certain number of persons. Had there been 20 more persons, each would have got ₹ 160 less. Find the original number of persons

34) In Figure, XY and X'Y' are two parallel tangents to a circle with centre O and another tangent AB with point of contact C intersecting XY at A and X'Y' at B. Prove that $\angle AOB = 90^\circ$



34) A train travels at a certain average speed for a distance of 63km and then travels at a distance of 72km at an average speed of 6km/hr more than its original speed. If it takes 3 hours to complete the total journey, what is the original average speed?

35) (a) If the mode of the following distribution is 55, then find the value of x . Hence, find the mean.

Class Interval	0 – 15	15 – 30	30 – 45	45 – 60	60 – 75	75 – 90
Frequency	10	7	x	15	10	12

OR

(b) A survey regarding heights (in cm) of 51 girls of class X of a school was conducted and the following data was obtained:

Heights (in cm)	Number of girls
less than 140	04
less than 145	11
less than 150	29
less than 155	40
less than 160	46
less than 165	51

Find the median height of girls. If mode of the above distribution is 148.05, find the mean using empirical formula.

(Section – E)

Section E consists of 3 case study-based questions of 4 marks each.

36) In a class, the teacher asks every student to write an example of A.P. Two boys Aryan and Roshan writes the progression as $-5, -2, 1, 4, \dots$ and $187, 184, 181, \dots$ respectively. Now the teacher asks his various students the following questions on progression.

Help the students to find answers for the following:

- Find the sum of the common difference of two progressions.
- Find the 34th term of progression written by Roshan.
- (A) Find the sum of first 10 terms of the progression written by Aryan.

OR

(B) Which term of the progressions will have the same value?

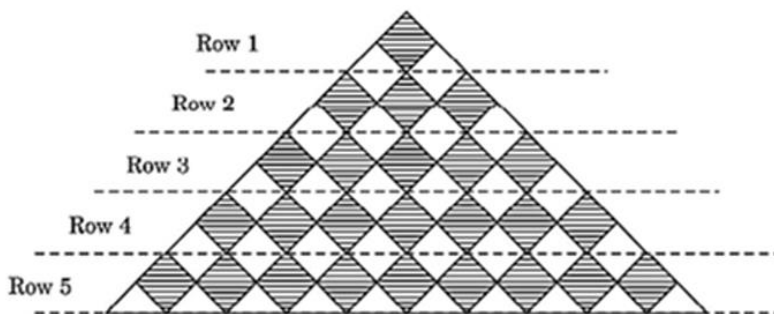
37) India gate (formerly known as All India war memorial) is located near Karthavya path. (formerly Rajpath) at New Delhi. It stands as a memorial to 74187 soldiers of Indian Army, who gave their life in the first world war. This 42m tall structure was designed by Sir Edwin Lutyens in the style of Roman triumphal arches. A student Shreya of height 1 m visited India Gate as a part of her study tour.



- i. What is the angle of elevation from Shreya's eye to the top of India Gate, if she is standing at a distance of 41m away from the India Gate?
- ii. If Shreya observes the angle of elevation from her eye to the top of India Gate to be 60° , then how far is she standing from the base of the India Gate?
- iii. (A) If the angle of elevation from Shreya's eye changes from 45° to 30° , when she moves some distance back from the original position. Find the distance she moves back. O

38) Read the text carefully and answer the questions:

A fashion designer is designing a fabric pattern. In each row, there are some shaded squares and unshaded triangles.



- a) Identify A.P. for the number of squares in each row.
- (b) Identify A.P. for the number of triangles in each row

OR

Write a formula for finding total number of triangles in n number of rows. Hence, find S_{10} .

- (c) If each shaded square is of side 2 cm, then find the shaded area when 15 rows have been designed